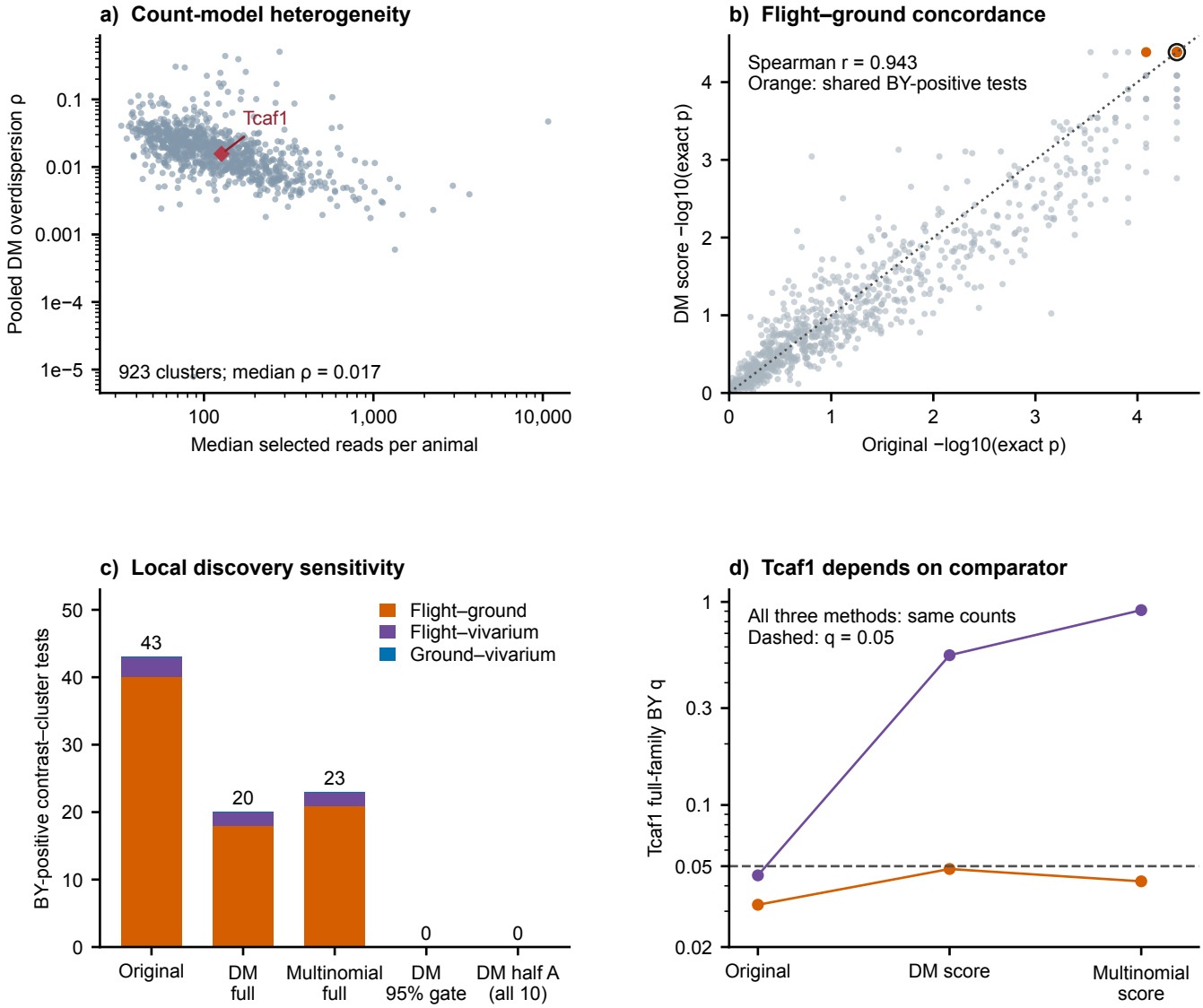

a) Count-model heterogeneity
Pooled DM overdispersion ρ
0.1
0.01
0.001
1e−4
1e−5
Tcaf1
923 clusters; median ρ = 0.017
100
1,000
10,000
Median selected reads per animal
b) Flight–ground concordance
Spearman r = 0.943
Orange: shared BY-positive tests
DM score −log10(exact p)
Original −log10(exact p)
0
1
2
3
4
c) Local discovery sensitivity
BY-positive contrast–cluster tests
Flight–ground
Flight–vivarium
Ground–vivarium
43
20
23
0
0
50
40
30
20
10
0
Original
DM full
Multinomial full
DM 95% gate
DM half A (all 10)
d) Tcaf1 depends on comparator
Tcaf1 full-family BY q
All three methods: same counts
Dashed: q = 0.05
1
0.3
0.1
0.05
0.02
Original
DM score
Multinomial score